

Mathematics

A Cultural Heritage

Lecture 2

Numbers – Part 2

Written Mathematics

Written Mathematics

- Egyptian Numbers:
 - 1 = Single stroke
 - 10 = Heel bone or cattle yoke
 - 100 = Piece of string
 - 1000 = Water Lily
 - 10000 = Finger
 - 100000 = Frog
 - 1 million = Slaves bowing

	=1
	=10
	=100
	=1,000
	=10,000
	=100,000
	=1,000,000

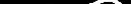
Egyptian Number System

Video excerpt from [The story of 1](#) (11:00-12:45)



Egyptian Number System

- Examples:

2342= 

10,463 = 10,000 + 400 + 60 + 3

$$1/5 = \frac{\text{1 diamond}}{\text{5 vertical lines}}$$
$$1/360 = \frac{\text{one lens}}{3 \text{ lenses} \times 3 \text{ lenses} \times 3 \text{ lenses}}$$

Egyptian Number System

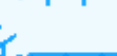
	1	2	3	4	5	6	7	8	9
UNITS									
TENS									
HUNDREDS									
THOUSANDS									
TENS OF THOUSANDS									
HUNDREDS OF THOUSANDS									

Egyptian hieratic numerals of the New Kingdom (1600–1000BC) used more symbols than before, making numbers more compact but harder to learn to use.

Babylonian Writing



Babylonian Number System

1 	11  	21  	31  	41  	51  
2 	12  	22  	32  	42  	52  
3 	13  	23  	33  	43  	53  
4 	14  	24  	34  	44  	54  
5 	15  	25  	35  	45  	55  
6 	16  	26  	36  	46  	56  
7 	17  	27  	37  	47  	57  
8 	18  	28  	38  	48  	58  
9 	19  	29  	39  	49  	59  
10 	20 	30 	40 	50 	

Babylonian Number System

- Hexagesimal: Groups of 60

- For us: $11 = 10 + 1$
- For them:   = $60 + 1$

13   

47  

$$\underbrace{3 \times 60^1}_{\text{from 180}} + \underbrace{28 \times 60^0}_{\text{from 28}} = 208$$

Babylonian Number System

A very large number:



MS 2351

Extremely large 15-place sexagesimal number. Babylonia, 19th c. BC

This is 104,857,600,000,000,000,000,000
in our decimal system!

<http://www.schoyencollection.com/math.html>

Ancient Greek Numbers

Letters used to represent numbers

α	alpha	1	ι	iota	10	ρ	rho	100
β	beta	2	κ	kappa	20	σ	sigma	200
γ	gamma	3	λ	lambda	30	τ	tau	300
δ	delta	4	μ	mu	40	υ	upsilon	400
ε	epsilon	5	ν	nu	50	φ	phi	500
ς	stigma *	6	ξ	xi	60	χ	chi	600
ζ	zeta	7	ο	omicron	70	ψ	psi	700
η	eta	8	π	pi	80	ω	omega	800
θ	theta	9	Ϟ	koppa *	90	Ϡ	sampi *	900

* The characters stigma, koppa, and sampi are now obsolete

Ancient Greek Numbers

- Examples:

- 35 would be: ΛE or $\lambda \epsilon$ = 30 + 5
- 204 would be $\Sigma \Delta$ or $\sigma \delta$ = 200 + 4

- Larger Numbers:

'Α	'Β	'Γ	'Δ	'Ε	'Ϛ	'Ζ	'Η	'Θ
1000	2000	3000	4000	5000	6000	7000	8000	9000

Source: www-history.mcs.st-and.ac.uk/HistTopics/Greek_numbers.html

Roman Numerals

I

one

V

five

X

ten

L

fifty

C

one hundred

D

five hundred

M

one thousand

Roman Numerals

1 I	21 XXI	41 XLI	61 LXI	81 LXXXI
2 II	22 XXII	42 XLII	62 LXII	82 LXXXII
3 III	23 XXIII	43 XLIII	63 LXIII	83 LXXXIII
4 IV	24 XXIV	44 XLIV	64 LXIV	84 LXXXIV
5 V	25 XXV	45 XLV	65 LXV	85 LXXXV
6 VI	26 XXVI	46 XLVI	66 LXVI	86 LXXXVI
7 VII	27 XXVII	47 XLVII	67 LXVII	87 LXXXVII
8 VIII	28 XXVIII	48 XLVIII	68 LXVIII	88 LXXXVIII
9 IX	29 XXIX	49 XLIX	69 LXIX	89 LXXXIX
10 X	30 XXX	50 L	70 LXX	90 XC
11 XI	31 XXXI	51 LI	71 LXXI	91 XCI
12 XII	32 XXXII	52 LII	72 LXXII	92 XCII
13 XIII	33 XXXIII	53 LIII	73 LXXIII	93 XCIII
14 XIV	34 XXXIV	54 LIV	74 LXXIV	94 XCIV
15 XV	35 XXXV	55 LV	75 LXXV	95 XCV
16 XVI	36 XXXVI	56 LVI	76 LXXVI	96 XCVI
17 XVII	37 XXXVII	57 LVII	77 LXXVII	97 XCVII
18 XVIII	38 XXXVIII	58 LVIII	78 LXXVIII	98 XCVIII
19 XIX	39 XXXIX	59 LIX	79 LXXIX	99 XCIX
20 XX	40 XL	60 LX	80 LXXX	100 C

200 CC 300 CCC 400 CD 500 D 600 DC 700 DCC 800 DCCC 900 CM 1000 M

Roman Numerals

- Roman numerals still appear



©Walt Disney Pictures

Roman Arithmetic

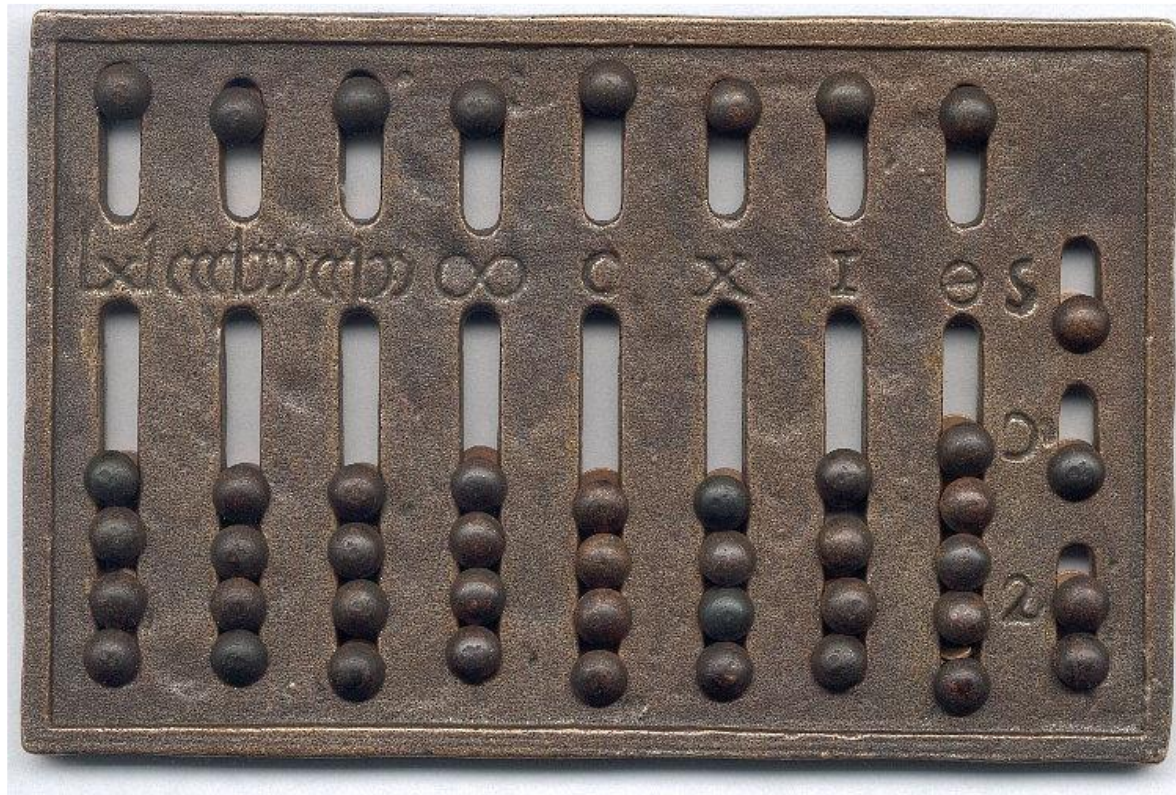
- **IV** is 4, **IX** is 9, but **XLIX** (not IL) is 49.
- Have to remember order of subtracting/adding
- Arithmetic difficult:

XXXVI +

XIX
LV

Roman Arithmetic

- Done with an **abacus**



Chinese Number System

四十 = $4 \times 10 = 40$

but 十四 $10 + 4 = 14$

and 四十四 $4 \times 10 + 4 = 44$

0	〇	零			
1	一	壹	10	十	拾
2	二	貳	20	廿/廿	貳拾
3	三	叁	30	卅	參拾
4	四	肆	40	卌	肆拾
5	五	伍			
6	六	陸	10^2	百	佰
7	七	柒	10^3	千	仟
8	八	捌	10^4	万	萬
9	九	玖	10^8	亿	億

Hindu – Arabic Number System

Brahmi	↓		—	=	≡	+	μ	Ϸ	7	5	7
Hindu	↓	०	१	२	३	४	५	६	७	८	९
Arabic	↓	•	١	٢	٣	٤	٥	٦	٧	٨	٩
Medieval	↓	0	I	2	3	℞	ç	6	Λ	8	9
Modern		0	1	2	3	4	5	6	7	8	9